Exp No:13 Experiment to handle missing values and inappropriate data in a given data set

AIM:

To understand and handle missing values and inappropriate data in a given data set.

CODE:

```
from google.colab import files
files.upload()
import numpy as np
import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")
df

# Identify duplicate row
df.duplicated()

# Display information
df.info()

# Remove duplicate rows
df.drop_duplicates(inplace=True)
df

# Length of dataframe
len(df)

# Reset index
index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
df

# Remove duplicate Age_Group column
df.drop(['Age_Group.1'], axis=1, inplace=True)
df

# Replace inappropriate values with NaN
df.CustomerID.loc[df.CustomerID < 0] = np.nan
df.Bill.loc[df.Bill < 0] = np.nan
df.EstimatedSalary.loc[df.EstimatedSalary < 0] = np.nan
df['NoOfPax'].loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20)] = np.nan

df
```

```
# Check unique values
df.Age_Group.unique()
df.Hotel.unique()

# Correct inappropriate hotel name
df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)

# Correct food preference values
df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
df.FoodPreference.replace(['non-Veg'], 'Non-Veg', inplace=True)

# Fill missing values
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)
df.NoOfPax.fillna(round(df.NoOfPax.median()), inplace=True)
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
df.Bill.fillna(round(df.Bill.mean()), inplace=True)

df
```

OUTPUT:

```
            <class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     object
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     object
 4   FoodPreference   11 non-null     object
 5   Bill             10 non-null     float64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group.1      11 non-null     object
```

```
df.Bill.fillna(round(df.Bill.mean()), inplace=True)
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	30-35	5	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	6	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	-1	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3	Ibis	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	7	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5	RedFox	Non-Veg	1801.0	4.0	87777.0

RESULT:

Thus,the given data was analyzed successfully and required output was obtained.